

Cao Tien Dung (Harry)

Senior Computer Systems Engineering Student | STEM & AI Systems Engineer

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PROFESSIONAL SUMMARY

Senior Computer Systems Engineering student at Da Nang University of Technology (GPA: 3.6/4.0; 2nd Prize in University-level Scientific Research in Applied AI). Expertly bridges rigorous STEM foundations (Advanced Mathematics, Embedded Systems) with end-to-end AI training, code benchmarking, and mathematical document authoring in LaTeX. Proven track record as a multidisciplinary specialist (STEM, LaTeX, and AI Evaluation) with strong capabilities in evaluating advanced LLMs, reasoning verification, and high-performance technical systems. Fluent in English (IELTS 6.5) and proficient in Japanese (JLPT N3).

TECHNICAL SKILLS

STEM Foundations & Math: Probability & Statistics, Linear Algebra, Calculus, Numerical Methods, Algorithms

Programming & Core: Python (Advanced), C/C++ (Systems & Memory Management), MATLAB, LaTeX (Scientific Publishing)

AI, Systems & Hardware: PyTorch, YOLOv8/v11, OpenCV, Grad-CAM++, STM32, ESP32, Altium Designer, PCB Design

Developer Tools: Git, GitHub Actions, Linux/Bash, Docker, FastAPI, SQLite, Technical Rubric Design

CORE EXPERTISE & EXPERIENCE

Multidisciplinary AI Trainer & Technical Specialist

Global AI Training Platforms & Enterprise Evaluation Frameworks

Remote

2024 – Present

- STEM Specialist (Sciences & Engineering):** Evaluated and solved complex technical problems spanning engineering, advanced mathematics, physics, and programming logic to train next-generation AI reasoning models.
- LaTeX Specialist Scientific Document Authoring:** Formulated, reviewed, and formatted complex mathematical equations, algorithms, and technical reports using advanced LaTeX formatting and structural design.
- Code Evaluation Comparative Benchmarking (Claude-Bench / CodeEval):** Designed test suites and execution harnesses to evaluate accuracy, algorithmic complexity, and instruction-following fidelity of code generation models (Claude Code, GPT) using strict grading rubrics.

KEY PROJECTS

DeepGuard — Deepfake Detection Platform | *PyTorch, EfficientNet-B4, Grad-CAM++*

2026

- Implemented deepfake detection framework using EfficientNet-B4; extensively utilized Grad-CAM++ for saliency mapping to diagnose failure modes and eliminate false positives.

DeepLock Pro — Biometric Face Security | *OpenCV, EAR, AES-128, CLAHE*

2026

- Engineered liveness detection (blink/pose estimation) to prevent spoofing attacks, securing 128-dim facial embeddings in AES-128 encrypted databases.

TrafficAI — Smart Vehicle Monitoring & LPR | *YOLOv8, OpenVINO, FastAPI*

2025 – 2026

- Built smart parking solution processing real-time IP camera feeds, optimizing YOLOv8 and EasyOCR models with Intel OpenVINO for low-latency edge inference.

FireGuard IoT — Smart Fire Detection | *ESP32, YOLOv8, Telegram API*

2025 – 2026

- Combined IoT sensors with YOLOv8 vision models, implementing temporal filtering to reduce false alarms by 90% and adding automated Telegram dispatching.

AI Automated Sorting Conveyor | *PBL4 | STM32, ESP32, OpenCV*

2025

- Developed STM32 OOP firmware for motor synchronization and integrated OpenCV barcode/QR scanning with Firebase real-time database logging.

OCL Audio Power Amplifier Hardware Design | *Altium Designer, Proteus*

2024 – 2025

- Designed and soldered a Hi-Fi analog audio power amplifier PCB using Altium Designer and Proteus simulation, performing oscilloscope waveform testing.

Tour Booking & E-Commerce Simulator | *C++, OOP, Data Structures*

2022 – 2023

- Engineered a console-based tour booking and Shopee-style e-commerce simulator applying advanced C++ OOP concepts and memory management.

EDUCATION & HONORS

Da Nang University of Technology (DUT)

Da Nang, Vietnam

Engineer's Degree in Computer Systems Engineering (Cumulative GPA: 3.6 / 4.0)

Mar. 2022 – Sep. 2027 (Expected)

- Award:** 2nd Prize in University-level Scientific Research in Applied Artificial Intelligence.
- Languages:** Vietnamese (Native), English (IELTS 6.5 – Advanced technical reading), Japanese (JLPT N3).

CERTIFICATIONS

AI & Machine Learning: Deep Learning Specialization (Coursera / DeepLearning.AI), PyTorch for Beginners (Udemy)

Programming & Tools: C/C++ (Codelearn), Python (Coursera), Advanced LaTeX Document Processing